Leakage-Aware Purchase Conversion Prediction in Sponsored Search

Research Design and Feasibility Protocol

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1. Research problem and contribution

Sponsored-search advertisers use post-click purchase probabilities to allocate budgets and assess the value of traffic. A predictive system must estimate whether a click will result in a purchase using information available when the click occurs. Retrospective logs, however, mix click-time attributes with outcomes observed later. A model can consequently achieve excellent offline scores yet be unusable at the proposed decision point. This study asks: How do feature-availability restrictions and temporal evaluation change the apparent discrimination and calibration of purchase-conversion models in sponsored search?

Three subquestions guide the research: (RQ1) Which logged variables are outcome-derived, potentially post-click, or plausibly available at click time? (RQ2) How robust is a regularized logistic-regression baseline under chronological evaluation and conservative predictor ablation? (RQ3) Do additional model complexity and probability calibration improve results once the same defensible feature set and evaluation protocol are used? The intended audience is researchers in applied machine learning, advertising analytics, and practitioners evaluating offline model reliability.

The anticipated contribution is a transparent, reproducible audit-and-evaluation protocol, not a claim to invent a new conversion algorithm. The paper will distinguish observed discrimination from decision-time validity, document sources of uncertainty, and report how conclusions change under alternative feature sets and label-maturity assumptions.

2. Scholarly evidence and research gap

Tallis and Yadav (2018) show that sponsored-search conversion prediction must adapt to variations in product demand, grounding the application and motivating temporal checks. McMahan et al. (2013) demonstrate why sparse, regularized linear models remain operationally meaningful in advertising; however, their primary task is click-through prediction rather than post-click purchase prediction. These studies justify a logistic baseline but do not establish that this particular retrospective sample can support a click-time conversion claim.

Kaufman, Rosset, and Perlich (2011) formalize target leakage as the use of information illegitimately available for prediction. Their framework makes a column-by-column availability audit necessary; merely dropping the target is insufficient. Chapelle (2014) separately demonstrates that delayed conversions complicate negative labeling because an unconverted click can still convert later. These are distinct threats: feature leakage concerns predictor timing, while label maturity concerns when ground truth becomes knowable.

Cerqueira, Torgo, and Mozetič (2020) find that evaluation schemes can behave differently under temporal nonstationarity, although their empirical subject is time-series forecasting rather than click-level classification. Davis and Goadrich (2006), together with Saito and Rehmsmeier (2015),

explain why precision-recall analysis is informative under class imbalance. Niculescu-Mizil and Caruana (2005) and Guo et al. (2017) establish that strong ranking performance does not guarantee reliable probabilities, motivating calibration checks.

Together, these sources provide well-established principles, but they do not jointly establish the sensitivity of this Criteo benchmark to predictor timing, chronological splits, and calibration. The evidence-based gap is therefore a bounded benchmark-specific synthesis: the study will test whether an apparently outstanding conversion score survives progressively stricter, explicitly documented evaluation assumptions. Novelty is methodological integration and reproducibility, not a universal claim about all sponsored-search systems.

3. Dataset, population, and measurement

The public Criteo Sponsored Search Conversion Log is an anonymized, sampled record of approximately 90 days of sponsored-search traffic. Its unit of analysis is a product-ad click, not a unique customer. The target, Sale, indicates whether the click was associated with a purchase within the dataset's 30-day conversion window (Criteo AI Lab, 2018). Predictors include price, hashed product attributes, device, audience, seller and user identifiers, click time, and a weekly-click field; hashed values are not interpretable as demographic characteristics.

Two locally executed notebooks provide feasibility evidence from a reproducible 500,000-click random sample drawn while scanning 15,995,634 records. The sample contains 54,264 conversions (10.8528%) and covers August 3–November 2, 2020 (UTC). Exploratory checks found 12 exact duplicate rows; product_category7 was entirely the -1 sentinel, while categories 6 and 5 had approximately 98.26% and 91.26% sentinel values. Missing-value markers will be distinguished from pandas nulls and evaluated for informative missingness. The sample is not presumed to represent an independently sampled production stream.

Crucially, SalesAmountInEuro has a -1 sentinel for all observed non-conversions and a non-sentinel value for all conversions in this sample; time_delay_for_conversion likewise closely tracks the outcome. Both are excluded from valid click-time predictors. nb_clicks_1week, audience assignment, and any updated product fields require provenance review. click_timestamp is used to define splits; it is not included in the current model feature matrix. Ethical safeguards include no user re-identification, no claims about protected groups from hashes, no publication of row-level logs, and compliance with Criteo terms of use.

4. Methods and evaluation

The initial benchmark uses a constant-probability comparator and L2-regularized logistic regression. The modeling notebook sorts sampled records by valid click timestamp and divides them into the earliest 60% for training, next 20% for validation, and latest 20% for testing. This yields 300,000/100,000/100,000 rows, with conversion prevalence of 11.36%/10.27%/9.91%, respectively. Missing-value handling, scaling, and categorical encoding are fitted on training data and then applied without refitting to later partitions. Model selection uses validation data; the test partition is reserved for final reporting.

The executed retrospective baseline produced validation average precision (AP) 0.9545 and test AP 0.9553, test ROC-AUC 0.9881, log loss 0.0488, and Brier score 0.0108; the constant predictor's test AP was 0.0991. These are preliminary observed scores, not proof of valid real-time conversion

prediction. Their magnitude is itself a motivation for an expanded predictor-availability investigation. No attribution to a particular remaining feature is justified before ablation.

Planned experiments will compare (A) a deliberately leakage-prone diagnostic model, labeled unequivocally invalid for deployment; (B) the current baseline excluding the two outcome-derived fields; and (C) a conservative subset excluding uncertain timing-dependent features. A histogram-based gradient-boosting model will be compared with logistic regression using identical splits and defensible inputs. Sensitivity experiments will vary feature groups and evaluate rolling chronological windows; a nominal 30-day embargo will be examined, with explicit acknowledgement that a gap cannot reconstruct unknown historical training snapshots or fully resolve label maturity in a retrospective export.

AP is the primary discrimination metric because the positive class is roughly one tenth of the sample; ROC-AUC is supplementary. Log loss, Brier score, and reliability diagrams will assess probabilistic quality. Test-set prevalence and the AP of a constant score will be reported as context. Confidence intervals may be estimated by time-block bootstrap where feasible, rather than naive row resampling that ignores temporal dependence. Interpretation will prioritize effect sizes and stability, not a single maximum score.

5. Reproducibility, feasibility, and anticipated outputs

The GitHub repository will contain a README with setup/run instructions, pinned environment requirements, the two executed Jupyter notebooks, modular preprocessing utilities where needed, and a .gitignore excluding data/ and generated private artifacts. The local source file is CriteoSearchData, read as a headerless, tab-separated 23-column dataset. Reproducible sampling uses a fixed seed (530); the repository will document package versions, split definitions, feature lists, and metrics. The public data download URL and terms will be linked rather than redistributing the dataset. Repository URL: [INSERT PUBLIC GITHUB URL BEFORE SUBMISSION].

Week 10 delivers this protocol, exploratory leakage evidence, and an executable chronological baseline. By Week 12, the plan is to complete predictor documentation, ablations, gap/window sensitivity checks, at least one nonlinear comparator, and uncertainty summaries. Week 13 delivers a short conference-style paper, reproducible tables/figures, and a five-minute presentation. Expected artifacts include a feature-availability matrix, missingness summary, temporal-split table, performance/calibration comparison, and limitations statement.

Limitations include anonymized feature semantics, unknown exact derivation times, a 30-day attribution window, possible user/product repetition, sampling effects, distribution shifts, and limited external validity beyond the 2020 log. A notably high baseline score will be treated as a research question requiring falsification, not evidence of causally meaningful or deployment-ready predictions.

6. Contribution statement

This is an individual project. The student is responsible for research-question formulation, scholarly synthesis, local dataset preparation, exploratory analysis, model implementation, evaluation design, repository documentation, interpretation, writing, and presentation. No teammates are claimed. Any assistance used for editing or code review must be disclosed if required by course policy.

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